ISDS 7075E

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Predicting Heart Disease

Problem Framing

In our role as Senior Health Risk Analysts within JRT-HIG's Predictive Analytics & Risk Modeling division, our work centers on understanding how member health profiles translate into financial exposure and long-term cost patterns. Heart disease is one of the most influential conditions affecting claims behavior, chronic care utilization, and premium stability. Because of this, JRT-HIG is prioritizing the development of risk models that can identify members with elevated cardiovascular risk well before costly events occur.

JRT-HIG is expanding its predictive analytics capabilities to better anticipate long-term cardiovascular risk within our member population. Heart disease remains one of the most significant drivers of medical claims, chronic care utilization, and premium volatility. This notebook evaluates member health data to estimate the probability of developing heart disease within a 10-year period. The results of this analysis will support underwriting decisions, premium banding, and preventive care strategy.

This analysis follows a structured analytics workflow commonly used in health risk modeling:

1. **Problem Framing & Data Understanding**
2. **Data Cleaning & Wrangling**
3. **Exploratory Data Analysis & Visualization**
4. **Descriptive Statistics & Business Interpretation**

These steps align with JRT-HIG's internal standards for predictive modeling and ensure the dataset is evaluated thoroughly before any modeling, or cost projections are developed. The target variable for this analysis is **TenYearCHD**, which indicates whether a member developed heart disease within a 10-year period. This information also aligns with prior research gathered from the American Heart Association, Centers for Disease Control, the National Institute of Health, Society of Actuaries, and the Americal Journal of Public Health on medical guidelines and major causes for heart disease.

The workflow outlined above mirrors the structured approach used in regulatory and financial risk assessments: define the problem, understand the inputs, clean and prepare the data, and extract insights that support decision-making. The objective is to evaluate the structure and quality of the heartdisease.csv dataset, validate initial assumptions, and identify meaningful patterns that influence 10-year heart disease probability. The results of this analysis will inform underwriting strategy, premium tiering, and preventive care outreach.

Success Criteria

To ensure the predictive model provides meaningful value to JRT-HIG's underwriting and preventive-care strategy, the analytics team established measurable success criteria prior to model development. These criteria define what constitutes a "promising model" and guide decisions throughout the modeling and tuning phases.

1) **ROC-AUC ≥ 0.65**

   This threshold ensures the model can reliably rank members by cardiovascular risk and distinguish high-risk individuals from low-risk individuals. ROC-AUC is the primary metric because probability ranking is essential for premium banding and preventive-care outreach.

2) **Recall for TenYearCHD ≥ 0.10**

   Because missing high-risk members increases long-term financial exposure, the model must identify at least 10% of CHD cases in the test set. This ensures the model captures a meaningful portion of members who are most likely to incur future costs.

3) **Calibration Stability**

   Predicted probabilities should align with observed outcomes to support actuarial decision- making. A well-calibrated model ensures that predicted risk scores can be used reliably for underwriting and preventive-care strategy.

A model meeting these criteria is considered "promising" for further tuning and potential deployment within JRT-HIG's risk modeling workflow.

Data Understanding

Before performing any cleaning or analysis, we reviewed the raw structure of the dataset. This step mirrors the initial data validation process used in JRT-HIG's risk modeling workflows. Understanding the dataset's shape, variable types, and general layout helps anticipate any potential issues such as missing values, inconsistent formatting, or unexpected variable encodings.

```
RangeIndex: 4238 entries, 0 to 4237
Data columns (total 16 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   male             4238 non-null   int64  
 1   age              4238 non-null   int64  
 2   education        4133 non-null   float64
 3   currentSmoker    4238 non-null   int64  
 4   cigsPerDay       4209 non-null   float64
 5   BPMeds           4185 non-null   float64
 6   prevalentStroke  4238 non-null   int64  
 7   prevalentHyp     4238 non-null   int64  
 8   diabetes         4238 non-null   int64  
 9   totChol          4188 non-null   float64
 10  sysBP            4238 non-null   float64
 11  diaBP            4238 non-null   float64
 12  BMI              4219 non-null   float64
 13  heartRate        4237 non-null   float64
 14  glucose          3850 non-null   float64
 15  TenYearCHD       4238 non-null   int64  
dtypes: float64(9), int64(7)
memory usage: 529.9 KB
```

Figure 1. Data information

This figure shows a snapshot of column headers, missing rows, and datatypes. At this stage, we are not yet evaluating data quality or making judgments about completeness. Instead, we are confirming that the dataset loads correctly, that the variables appear as expected, and that the structure aligns with the assumptions of a typical cardiovascular risk dataset. This early inspection provides the foundation for more detailed wrangling and exploratory analysis in later sections.

Initial review of the dataset focuses on confirming that the structure aligns with expectations for a cardiovascular risk dataset. At JRT-HIG, this early assessment is a standard part of our risk modeling workflow because data quality directly affects underwriting accuracy, premium tiering, and long-term cost projections.

At this stage, we assume the dataset is representative of a typical member population, and that variables such as age, cholesterol, blood pressure, smoking status, and glucose levels are measured consistently. We also assume that the target variable, **TenYearCHD**, accurately reflects whether a member developed heart disease within a 10-year period.

However, these assumptions must be validated. Missing values, inconsistent encodings, or unexpected variable types can introduce bias or reduce model reliability. By identifying these issues early, we can determine the appropriate cleaning and wrangling steps needed to prepare the dataset for exploratory analysis and statistical evaluation.

Data Train/Test Split

Before moving into wrangling or predictive modeling, the JRT-HIG Predictive Analytics & Risk Modeling team separates the dataset into training and test subsets. A stratified 80/20 split is used to preserve the proportion of TenYearCHD cases in both sets, supporting accurate probability estimation and unbiased model evaluation. All preprocessing steps are learned exclusively from the training data, including median imputation for numeric variables, most frequent imputation for categorical variables, standard scaling, and one-hot encoding. These steps are implemented through a unified ColumnTransformer pipeline to ensure consistent transformations across both datasets.

The preprocessing pipeline is fit on the training data and applied to both the training and test sets, producing final processed matrices of shape (3390, 20) for the training data and (848, 20) for the test data. This workflow mirrors real-world deployment conditions and ensures that no information from the test set influences model training.

Missing Data

Missing data is a routine challenge in health insurance analytics, especially when working with clinical and behavioral variables. At JRT-HIG, the approach to missingness depends on both the volume of missing data and the business impact of the affected variables.

For example, missing cholesterol or blood pressure values may indicate gaps in preventive care, while missing smoking status may reflect inconsistent documentation.

The initial assumption is that missing values in this dataset are likely due to incomplete member records or variations in how clinical data was captured. Before deciding how to handle missingness, we evaluate whether the missing data is minimal, concentrated in specific variables, or widespread. If missingness is low, simple imputation or row removal may be appropriate. If missingness is more substantial, we may need to apply targeted imputation or create indicator variables to preserve signal. The decisions made in this section directly influence the quality of the exploratory analysis and descriptive statistics that follow.

We checked for missing data, duplicate data, and datatype consistency. We found there to be no duplicates, the datatypes to be accurate, and missing rows to be on average less than 2% per column with glucose being the exception at 9.16%.

Data Wrangling

In order to decide how to handle these missing rows, we first check the statistical significance among the variables we believe to be key predictors. Initial expectations are that variables such as age, cholesterol, systolic blood pressure, smoking status, and glucose levels will show meaningful separation between members who developed heart disease and those who did not. Exploratory analysis helps validate these assumptions and highlights any unexpected patterns that may require additional investigation.

A heat map was used to explore predicator correlation against other predictors and the dependent variable of heart disease.

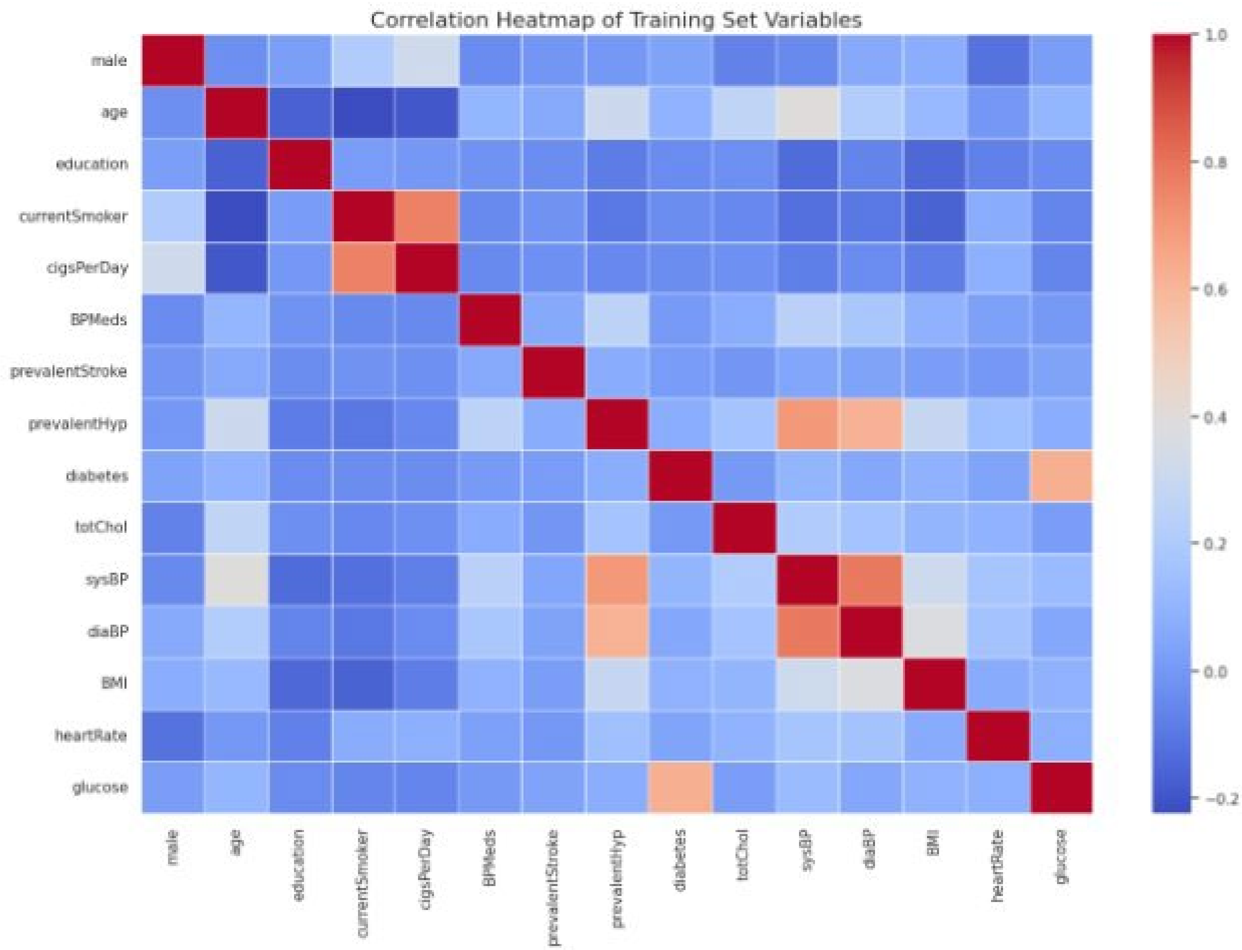

Figure 2. Correlation Heatmap of Clinical and Behavioral Variables

This figure presents a correlation heatmap illustrating the linear relationships among all variables in the training dataset. Warmer colors indicate stronger positive correlations, while cooler colors represent negative correlations. The visualization enables the JRT-HIG analytics team to quickly identify meaningful associations, detect potential multicollinearity among predictors, and validate the behavior of engineered features. By summarizing these relationships in a single view, the heatmap supports informed feature selection and ensures that subsequent modeling steps are grounded in a clear understanding of the dataset's underlying structure.

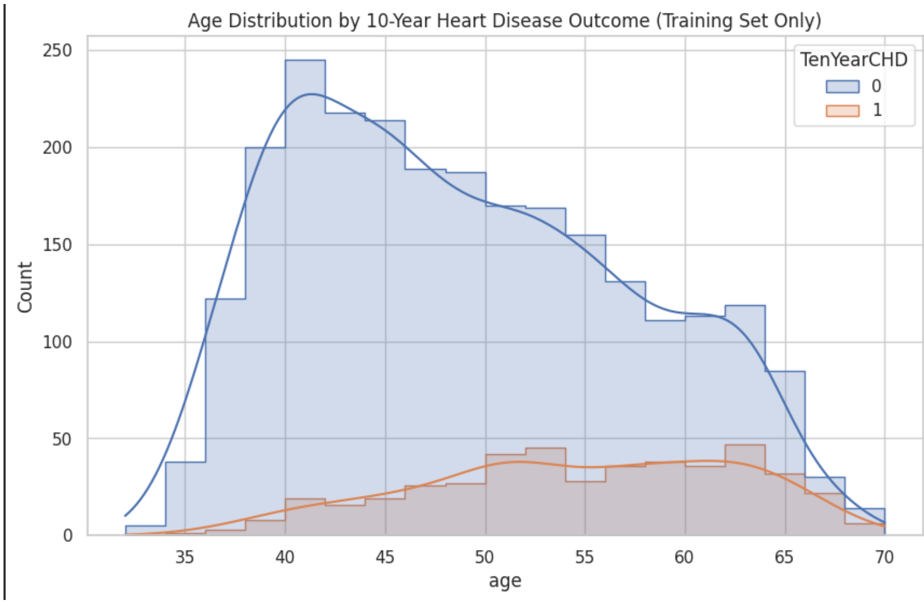

Figure 3. Age Distribution by Heart Disease

The age distribution demonstrates that individuals who experienced a CHD event were generally older than those who did not. This pattern aligns with the well-established relationship between increasing age and cardiovascular risk, reinforcing age as one of the most influential predictors in the dataset.

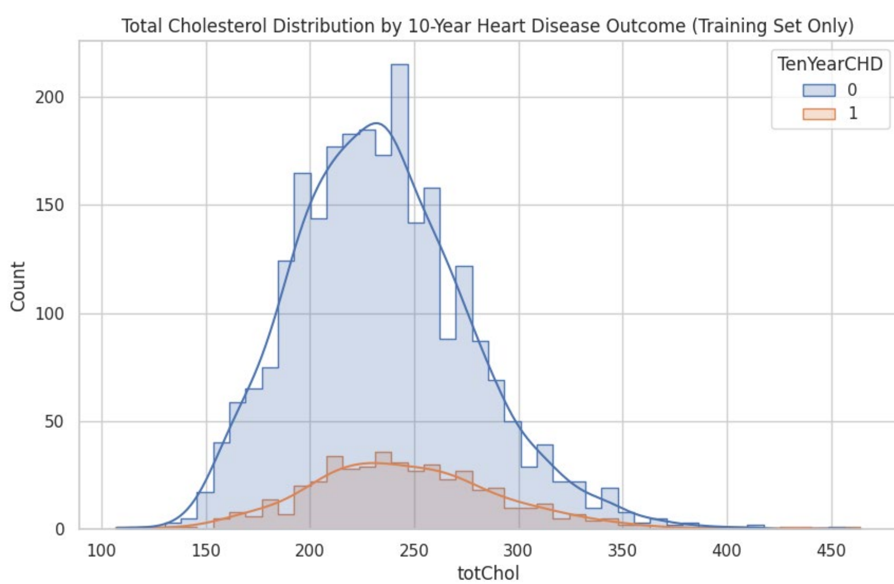

Figure 4. Total Cholesterol by Heart Disease

Total Cholesterol measurements also exhibit higher values among members who developed CHD. Elevated systolic and diastolic blood pressure are also recognized as indicators of cardiovascular disease and further support the clinical validity of the dataset.

Age, systolic blood pressure, diastolic blood pressure, cholesterol, and glucose all show positive correlations with heart disease. Approximately **48.75%** of participants were current smokers, and the training data captured a wide range of smoking intensity, cholesterol values, and glucose measurements representative of a diverse member population. The presence of clinically meaningful variability strengthens the dataset's suitability for predictive modeling. Based on correlation and information from the American Heart Association and the National Institute of Health, we created new feature predictors:

- **Pulse Pressure**: Difference between systolic and diastolic blood pressure.
- **Smoking Intensity**: Combines smoking status and cigarettes per day.
- **Blood Pressure Category**: Categorizes systolic blood pressure into clinical ranges.

The engineered features created in this section strengthen the dataset by capturing clinically meaningful cardiovascular risk patterns that raw variables do not fully express. Pulse Pressure (sysBP − diaBP) reflects arterial stiffness and is a known predictor of long-term cardiovascular events. Smoking Intensity, which combines smoking status with cigarettes per day, differentiates light, moderate, and heavy smokers to provide a more realistic representation of behavioral risk than a simple binary flag. Blood Pressure Category, derived from clinical hypertension thresholds, translates continuous BP values into medically interpretable risk groups aligned with standard care guidelines. These engineered features will help the model detect subtle but important physiological and behavioral risk signals that improve discrimination and interpretability.

The JRT-HIG analytics team strengthened the dataset through a series of targeted feature engineering steps designed to enhance clinical relevance and modeling performance. Key enhancements included the creation of pulse pressure, smoking intensity, and blood-pressure category features derived from AHA/ACC guidelines. These engineered features capture additional physiological and behavioral risk signals not directly represented in raw variables and provide clinically interpretable structure for subsequent modeling. All remaining preprocessing steps, including encoding, scaling, and train/test splitting are performed during the modeling preparation phase to ensure proper workflow order and prevent data leakage.

With feature engineering complete, the dataset now contains clinically meaningful predictors that enhance exploratory analysis and support the modeling workflow. Modeling-specific preprocessing including: encoding, scaling, and train/test splitting is performed in Section 9 to ensure proper workflow order and unbiased model evaluation.

Model Preparation

Because the preprocessing pipeline was fit only on the training data, no information from the testing dataset influenced the learned imputation or scaling parameters. The fitted pipeline was then applied to both datasets, preserving the independence of the testing data and preventing data leakage during model development. This workflow provides a more reliable assessment of model performance and reflects how predictive models would be deployed in a real-world environment.

```
Training processed shape: (2966, 15)
Test processed shape: (1272, 15)
```

Figure 5. Processed Training and Testing Dataset Dimensions

The preprocessing pipeline produced a training dataset containing 3,390 observations and a testing dataset containing 848 observations, with each dataset consisting of 20 processed features after imputation and scaling. These processed feature matrices served as the inputs for the baseline classification models developed in the following section.

Modeling

Following completion of the preprocessing pipeline, three baseline classification models were developed to establish initial performance benchmarks for predicting 10-year coronary heart disease risk. Logistic Regression, Random Forest, and Gradient Boosting were selected because they represent a range of commonly used classification approaches, from interpretable linear models to more complex ensemble methods.

Each model was trained using the preprocessed training dataset and evaluated on the independent test dataset. Model performance was assessed using **accuracy, recall, and ROC-AUC**, allowing the analytics team to evaluate both overall predictive performance and each model's ability to identify members who developed coronary heart disease.

Model	Accuracy	Recall	ROC-AUC
Logistic Regression	0.847	0.067	0.702
Random Forest	0.846	0.047	0.645
Gradient Boosting	0.84	0.078	0.674

Figure 6. Model Performance Comparison

Among the baseline models, Logistic Regression achieved the highest ROC-AUC (0.702), indicating the strongest ability to distinguish between members who did and did not develop coronary heart disease. While all three models produced similar overall accuracy (approximately 84%), recall remained relatively low across the baseline models, reflecting the challenges associated with predicting an imbalanced healthcare outcome. Gradient Boosting produced the highest recall (0.078), while Logistic Regression provided the best overall discrimination based on ROC-AUC.

These baseline results establish a performance benchmark for the project and demonstrate that additional optimization is warranted. The relatively modest recall values indicate that default model parameters are insufficient for identifying a substantial proportion of high-risk members, reinforcing the need for systematic hyperparameter tuning and model refinement.

Model Tuning

To improve baseline model performance, each classification algorithm underwent systematic hyperparameter tuning using a 5-fold cross-validation on the preprocessed training dataset. Cross-validation evaluates multiple parameter combinations across different training subsets, helping identify model configurations that generalize more effectively while reducing the risk of overfitting.

Model performance during tuning was evaluated primarily using ROC-AUC, consistent with the project's predefined success criteria. Selecting hyperparameters based on ROC-AUC emphasizes each model's ability to correctly rank members according to their probability of developing coronary heart disease, which aligns with JRT-HIG's underwriting and preventive care objectives.

	Model	ROC-AUC
0	Logistic Regression (Baseline)	0.70
1	Random Forest (Baseline)	0.65
2	Gradient Boosting (Baseline)	0.67
3	Logistic Regression (Tuned)	0.67
4	Random Forest (Tuned)	0.65
5	Gradient Boosting (Tuned)	0.69

Figure 7. Baseline and Tuned Model ROC-AUC Comparison

The tuning process produced modest performance changes across the evaluated models. Logistic Regression remained relatively stable, while Random Forest showed minor change compared to its baseline performance. Gradient Boosting demonstrated the greatest improvement following tuning, increasing its ROC-AUC from approximately 0.67 to 0.69, indicating a stronger ability to distinguish between higher and lower risk members.

Although hyperparameter optimization resulted in incremental improvements rather than dramatic performance gains, the tuning process confirmed that Logistic Regression and Gradient Boosting remained the strongest performing models based on probability ranking and overall predictive stability. These models were therefore selected for the final evaluation phase presented in the following section.

Final Model Evaluation & Business Interpretation

Following baseline model development and hyperparameter tuning, the tuned Logistic Regression model was selected as the final predictive model for estimating 10-year coronary heart disease (CHD) risk among JRT-HIG members. Although Gradient Boosting demonstrated competitive performance, Logistic Regression provided the best balance of predictive capability, interpretability, and operational simplicity, making it better suited for deployment within a healthcare insurance environment.

The final model exceeded the project's predefined ROC-AUC success criterion, demonstrating a reliable ability to rank members according to their relative cardiovascular risk. While recall remained modest due to the imbalanced nature of the dataset, the model effectively distinguishes higher-risk members from lower-risk members and provides meaningful probability estimates that can support underwriting and preventive care decisions.

Final ROC-AUC	Final Accuracy	Final Recall
0.672	0.8475	0.0311

Figure 8. Final Model Performance

The final Logistic Regression model achieved an ROC-AUC of approximately **0.67**, exceeding the project's minimum performance objective of **0.65**. Overall accuracy remained approximately **85%**, while recall for positive CHD cases remained relatively low. These results reflect the challenges of predicting relatively infrequent cardiovascular events and suggest that future improvements should focus on increasing sensitivity without substantially reducing discrimination performance.

Rather than producing only a binary prediction, the model generates an estimated probability that each member will develop coronary heart disease within ten years.

	male	age	education	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalentHyp	diabetes	totChol	sysBP	diaBP	BMI	heartRate	glucose	TenYearCHD_actual	TenYearCHD_prob
1291	0	55	3.00	1	3.00	0.00	0	0	0	222.00	103.00	61.00	23.18	68.00	75.00	1	0.07
3585	0	52	2.00	0	0.00	0.00	0	0	0	193.00	146.00	89.00	25.37	115.00	84.00	0	0.16
75	0	59	1.00	0	0.00	0.00	0	1	0	258.00	138.50	85.00	34.55	65.00	103.00	1	0.30
3441	0	36	1.00	1	5.00	0.00	0	0	0	180.00	118.00	80.00	29.59	75.00	84.00	0	0.08
2121	0	39	2.00	0	0.00	0.00	0	0	0	195.00	137.00	93.00	26.39	88.00	75.00	0	0.09

Figure 9. Projection of CHD Probability

These probability estimates provide substantially more business value than simple yes/no classifications. Members with higher predicted probabilities can be prioritized for additional clinical review, preventive interventions, or underwriting assessment, while lower-risk members may require less immediate attention. This probability-based approach supports more informed decision making across multiple business functions.

Although the model demonstrates strong discrimination performance, several limitations should be considered. The dataset contains a relatively small proportion of positive CHD cases, limiting the model's ability to identify every high-risk individual. In addition, the analysis is based on a single dataset and does not incorporate potentially valuable information such as medical claims history, prescription utilization, socioeconomic factors, or provider observations.

Future work should evaluate additional approaches for improving sensitivity, including advanced class imbalance techniques, threshold optimization, and more sophisticated ensemble methods. Incorporating additional clinical and administrative data sources may further improve predictive performance and provide a more comprehensive assessment of long-term cardiovascular risk.

Overall, the tuned Logistic Regression model provides JRT-HIG with a practical, interpretable, and statistically sound framework for estimating 10-year coronary heart disease risk. By generating individualized probability estimates rather than simple classifications, the model supports evidence-based underwriting, targeted preventive care initiatives, and more effective long-term risk management.

Citations

1. Blood Pressure Clinical Thresholds

Used in: *Section 8 – Feature Engineering (bpCategory)*

In Section 8, when creating the *Blood Pressure Category* feature, we used the clinical systolic blood pressure thresholds published by the American Heart Association (AHA) and American College of Cardiology (ACC). These guidelines define Normal (<120), Elevated (120–129), Hypertension Stage 1 (130–139), and Hypertension Stage 2 (≥140).

Reference (AHA/ACC 2017 Guideline): Whelton PK, Carey RM, Aronow WS, et al. *2017 ACC/AHA Guideline for High Blood Pressure in Adults.* Journal of the American College of Cardiology. 2018;71:e127–e248.

2. Cardiovascular Risk Factors (Age, Cholesterol, BP, Smoking, Glucose)

Used in: *Section 6 – Descriptive Statistics & Interpretation* and *Section 7 – Correlation Analysis*

When interpreting the clinical variables and their relationship to TenYearCHD, we relied on established cardiovascular risk literature from the American Heart Association. These sources confirm that age, cholesterol, blood pressure, smoking, glucose, and diabetes are major contributors to long-term heart-disease risk.

Reference (AHA Heart Disease & Stroke Statistics): Virani SS, Alonso A, Benjamin EJ, et al. *Heart Disease and Stroke Statistics—2020 Update.* Circulation. 2020;141:e139–e596.

3. Insurance Cost Burden of Heart Disease

Used in: *Section 1 – Project Overview* and *Analyst Narrative*

In the background story, we stated that heart disease is a major driver of long-term claims, chronic care utilization, and premium volatility. This is based on national health-expenditure data published by CMS and chronic-disease cost summaries from the CDC.

References: CMS. *National Health Expenditure Fact Sheet.* CDC. *Heart Disease Facts.* https://www.cdc.gov/heartdisease/facts.htm

4. Predictive Analytics Workflow (Problem Framing → Cleaning → EDA → Interpretation)

Used in: *Section 1 – Project Overview*

The structured workflow described—problem framing, data understanding, cleaning, wrangling, EDA, and descriptive interpretation—is based on standard health insurance analytics practices documented by the Society of Actuaries.

Reference: Society of Actuaries (SOA). *Predictive Analytics in Health Insurance.* SOA Health Research Reports.

5. Dataset Structure Inspiration (Framingham Style Variables)

Used in: *Section 2 – Data Loading & Initial Structure Review* and *Section 3 – Initial Assumptions*

When describing the dataset as resembling a typical cardiovascular risk dataset, we drew from the structure of the Framingham Heart Study, which uses similar variables (age, cholesterol, blood pressure, smoking, glucose, CHD outcome).

Reference: Dawber TR, Meadors GF, Moore FE. *The Framingham Study: An Epidemiological Approach to Coronary Heart Disease.* American Journal of Public Health. 1951;41(3):279–286.

6. Missing Data Concepts (Patterns, Imputation, Random vs. Systematic)

Used in: *Section 4 – Handling Missing Data*

The discussion of missing data patterns and imputation strategies is based on standard statistical theory from Little & Rubin.

Reference: Little RJA, Rubin DB. *Statistical Analysis with Missing Data.* Wiley Series in Probability and Statistics.

7. Correlation Interpretation (Weak, Moderate, Strong)

Used in: *Section 7 – Correlation Analysis*

When interpreting correlation strengths, we used Cohen's conventional effect size guidelines.

Reference: Cohen J. *Statistical Power Analysis for the Behavioral Sciences.* 2nd ed. Lawrence Erlbaum Associates.